EDMS 10-400-1

SPECIFICATION

FOR

Supply of the medium Voltage

Disc Porcelain Insulators

Anti-Fog Type

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توجد بنود اختيارية (Option items) يجب تحديدها بواسطة شركة التوزيع قبل الطرح.

CONTENTS

1. SCOPE:

The purpose of this specification is the design, construction, manufacturing, testing, packing and supply of the medium voltage disc porcelain (anti-fog type) insulators which should be mounted in isolated medium voltage networks 11 kV, 22 kV

2. APPLICABLE STANDARDS

Unless otherwise specified in this specification, the offered insulators shall comply with all the latest relevant applicable standards as following:

ANSI C29.1 Test methods for electrical power insulators.

ANSI C29.2 Wet process porcelain and toughened glass insulators suspension type.

BS-EN-ISO1461 Hot dip galvanized coating on iron and steel articles. Also, previously as BS 729

IEC 60060 High voltage testing techniques -Part 1: General definitions and techniques

IEC 60071 -1 Insulation co-ordination -Part 1: Terms, definitions, principles and rules

IEC 60437 Radio interference test on high voltage insulators

IEC 60507 Artificial pollution tests on high voltage insulators to be used on AC systems.

IEC 60815 Selection and dimensioning of high-voltage insulators intended for use in polluted conditions - Part 1: Definitions, information and general principles.

3. ENVIROMENTAL CONDITIONS

Ambient temperature	: - 5°C to +45°C (50°C as option)
Maximum relative Humidity	: 90 %
Maximum altitude	: 1000 m
Salt air	: ≤ 0.3 mg/sq.cm
Service dust storm	: Dusty atmosphere

4. DESIGN CRITERIA

1. The insulator should be light in weight and made of homogeneous best quality glazed porcelain free from flaws, cracks, voids, conducting impurities and chopped edges.
2. The glaze should be uniform, tough, brown, in color and free from any kind of defects
3. The insulator should be a one-piece construction.
4. The material of insulator should be thoroughly vitrified so that the insulating property is independent upon the glaze.
5. the pin should occupy with zinc sleeve.
6. The designated size of ball coupling should be size 16A and the socket is of size 16A or16 B that according to IEC 60120 standard.
7. The ball and pin should be made of forged steel.
8. The cap should be made of malleable or ductile cast iron and the locking device should be made of corrosion-resistant metal.
9. The cap and pin should be heavily hot-dip galvanized that in accordance with IEC standards for zinc coating & fixed to the porcelain insulator by means of neat Portland cement.
10. The pin, ball, cap socket and looking device shall be constructed in such a manner to avoid accidental separation of insulator string during erection or service and to give possibility for easy interchanging of insulator.
11. The Disc insulator should be constructed in such a way that:

    a) It should be not affected during service by atmospheric conditions like salts, ozone, acids, alkaline, dust and sandy storm.
    b) The fixation of metallic parts (cap and pin) to the porcelain insulator should be executed in such a manner to eliminate absolutely the defects, which may take place as a reset different expansion coefficient of materials used especially at rapid changes of temperatures.

12. The design of insulator and the metallic parts should be such as to minimize local corona formations and discharges causing radio interferences.
13. The metallic parts of the insulator (cap and pin) should have a factor of safety not less than 2.5 times the elastic limits of their materials

5. TECHNICAL SPECIFICATION:

The electrical and mechanical characteristics for disc porcelain insulator should be as follows:

Disc type insulator:

a) One unit (disc):

No.	Characteristic	Value (Anti-fog type)
1	Rated voltage	11 kV, 22 kV
2	Nominal frequency	50 Hz
3	Dry power frequency flashover voltage	≥ 80 kV
4	Wet power frequency flashover voltage	≥ 48 kV
5	Dry power frequency withstand voltage	≥ 75 kV
6	Wet power frequency withstand voltage	≥ 42 kV
7	Impulse 1.2/50μs withstand voltage is not less than	120 kV
8	power frequency puncture voltage is not less than	130 kV
9	Min. Leakage distance	450 mm
10	Min. Diameter for the disc	280 mm
11	Minimum breaking load	9000 Kg

b) String:

15 minutes power frequency withstand voltage for a string of insulator, (Anti-Fog) of two discs polluted with conductivity not less than 50 micro-siemens not less than (kV)	9 for 11 kV
15 minutes power frequency withstand voltage for a string of insulator, (Anti-Fog) of three discs polluted with conductivity not less than 50 micro-siemens not less than (kV)	13.8 for 22 kV

6. INSPECTION AND TESTING:

1. ….EDC reserves its right to carry out inspection during manufacturing stages and to witness the routine tests of the insulators.
2. Unless otherwise mutually specified or approved in writing, selection of samples, testing and rejection of insulators should be in accordance to all relevant latest applicable IEC standards and test report certificate should be provided.
3. ……EDC shall - before taking over the insulators - carry out the required tests on random samples taken from each shipment (in the case of there is no approved laboratory for the supplier in Egypt) in the Extra High Voltage Test Laboratory of the Egyptian Electricity Holding Company, If the random samples fail to pass the tests, ...EDC will have the right to reject the insulators.
4. Type, sample and routine tests certificates should be submitted to the …..EDC.
5. Attendance of the .....EDC inspection engineer during tests shall not relieve the manufacturer from his full responsibility of furnishing the insulators in compliance with the requirements of this specification and shall not give him the right to invalidate any claim the ….EDC makes because of defective or unsatisfactory material or faulty workmanship .
6. The required type tests of the insulators should be:
   a) Visual inspection.
   b) Verification of dimensions.
   c) Temperature cycle test.
   d) Wet power frequency flashover voltage.
   e) Dry power frequency flashover voltage.
   f) Wet power frequency withstands voltage.
   g) Dry power frequency withstands voltage.
   h) Puncture test.
   i) Dry Lightning withstand voltage.
   j) Electromechanical failing load test.
   k) Artificial pollution test (polluted withstand voltage test).
   l) Galvanizing test.
   m) Porosity test.
   n) Verification of the axial and radial displacement.

7. The required routine/sample tests of the insulators should be:

   a. Visual inspection.
   b. Verification of dimensions.
   c. Temperature cycle test.
   d. Porosity test.
   e. Puncture test.
   f. Verification of the locking system.
   g. Verification of the axial and radial displacement.
   h. Electromechanical failing load test.
   i. Galvanizing test.

(The samples will be randomly collected by ….EDC representative from the LOT as per IEC383-1)

7. MARKING

Each insulator will be clearly, and inedibility marked with the following:

1- Electromechanical failing load.

2- Manufactures names / trade mark

3- Year of manufacture.

4- Name of the manufacturing country.

5- Type and model of insulator

8. PACKING:

The insulators should be packed in suitable wooden crates or cases to withstand shipment, loading, handling and storing.

9. DRAWING AND CATALOGUE

- All necessary drawings should be submitted with the offer.
- Technical catalogues containing the technical data for all offered components of the unit should be submitted.

10. GUARANTEE TABLES

- The tenderer should fill in the attached guarantee tables.
- Any tender not accompanied with clear and complete guarantee tables will be rejected.

GUARANTEE TABLE
FOR
MEDIUM VOLTAGE
DISC PORCELAIN INSULATORS

Anti-Fog TYPE

No.	Description	Unit	Particulars
1	General data		
1.1	Designation		
1.2	Country of origin		
1.3	Type		
1.4	Manufacture		
2	Constructional data		
2.1	Insulator material		
2.2	Electro mechanical falling load	KN	
2.3	Diameter	Mm	
2.4	Spacing	Mm	
2.5	Color of insulator		
2.6	Creepage distance	mm	
	Protected	mm	
	Un protected	mm	
2.7	Ball and socket coupling size IEC60120		

2.8	Weight /approx..	Kg	

2.9	Principle drawing		
3	<u>Technical data</u>		
3.1	Wet power frequency flash over voltage	kV	
3.2	Dry power frequency flash over voltage	kV	
3.3	Wet power frequency withstand voltage	kV	
3.4	dry impulse withstand voltage	kV	
3.5	50% impulse flashover positive	kV	
3.6	50% impulse flashover negative	kV	
3.7	Puncture voltage in oil	kV	
3.8	15 minutes power frequency withstand voltage for a string of insulator, (Anti-Fog) of two discs polluted with conductivity not less than 50 micro-siemens not less than (kV)	kV	
3.7	Quality Certificate of Manufacture		

I/ We guarantee the data given above for the equipment offered.

Signature: ……………………………

Date: …………………………